Curriculum for regression ICL: a clean / noisy trade-off

Review for Srikar Babu Gadipudi | Noah Lund Syrdal | Revised 12 October 2026

Main finding. Compared with the exact same batches shuffled in time, curriculum improves clean regression but is worse with noisy demonstrations in **19/20** seeds at SD1 and **16/20** at SD2. All models were trained on clean tasks. Noise results are secondary, descriptive comparisons at the final checkpoint.

Final evaluation (125k updates)	Curriculum mean	Shuffled mean	Curriculum vs shuffled
Clean	0.00773	0.01208	17/20 better
Gaussian label noise, SD1	0.08044	0.07245	19/20 worse
Gaussian label noise, SD2	0.32529	0.26421	16/20 worse

NMSE = clean-target squared error / 50, averaged over 50-100 demonstrations on 4,096 unseen tasks. Noise affects demonstration labels, not the query target. An independent final evaluation bank repeats these win/loss counts.

Ordering: smaller typical gains, sensitive mean

The **prespecified clean learning-curve** comparison has 16/20 wins (two-sided exact sign-test p = 0.01182), mean AUC reduction 0.02570 and median reduction 0.00748. Seed 1017 supplies 69.3% of the aggregate gain. Removing it descriptively gives mean reduction 0.00832, below the frozen 0.02 minimum. The original analysis retains all 20 seeds.

Final clean error is 0.665 times shuffled error using the geometric mean of paired ratios (0.688 without seed 1017). For **clean AUC**, the corresponding ratios are 0.607 and 0.685. Additional exact Wilcoxon signed-rank p-values on paired AUC differences are 0.000134 and 0.000267, respectively. Ratios, exclusions and Wilcoxon tests are **exploratory**; they do not replace the original sign test or its mean-effect requirement.

Full difficulty: a limited optimizer search

Full-difficulty final outcomes	Seeds	Mean NMSE	Median NMSE	Range
Learned (NMSE <= 0.25)	10/20	0.05093	0.02592	0.01319-0.17858
Did not learn	10/20	0.98336	0.99004	0.93861-0.99752

The overall mean 0.51715 falls between these groups. In development, 11/12 full-difficulty runs had AUC near 0.996. Its selected learning rate (0.0001) was the lowest on the grid; this is not a demonstrated strong or optimal baseline. Curriculum beats it in 20/20 paired AUCs (sign-test p = 0.00000191), but matched shuffled exposure is the more informative ordering control.

Design and prespecified analysis

Twenty paired seeds per strategy, 125k updates; 12-layer Transformer, hidden size 256, eight heads; batch 256. Curriculum grows dimension 5 to 50 and the number of supervised prediction positions over 25k updates. Shuffled uses the **identical realized batch multiset**, including targets, masks and scales, in permuted order. Both then share the same 100k full-difficulty tail. Physical sequence shapes and expected target variance stay fixed.

A separate 36-run development search selected the shared optimizer using shuffled performance; its setting ranked third for curriculum, making the choice conservative relative to curriculum development preferences. Full difficulty selected its own setting from that grid. The locally frozen protocol tested curriculum vs full difficulty first, then vs shuffled: each required sign-test $p < 0.05$, mean AUC reduction $>= 0.02$, and curriculum final mean $<= 0.25$. Both passed. AUC averages clean NMSE over updates 25k-125k.

All-seed evidence and interpretation

Robust summaries retain every seed; full-difficulty runs are shown individually.

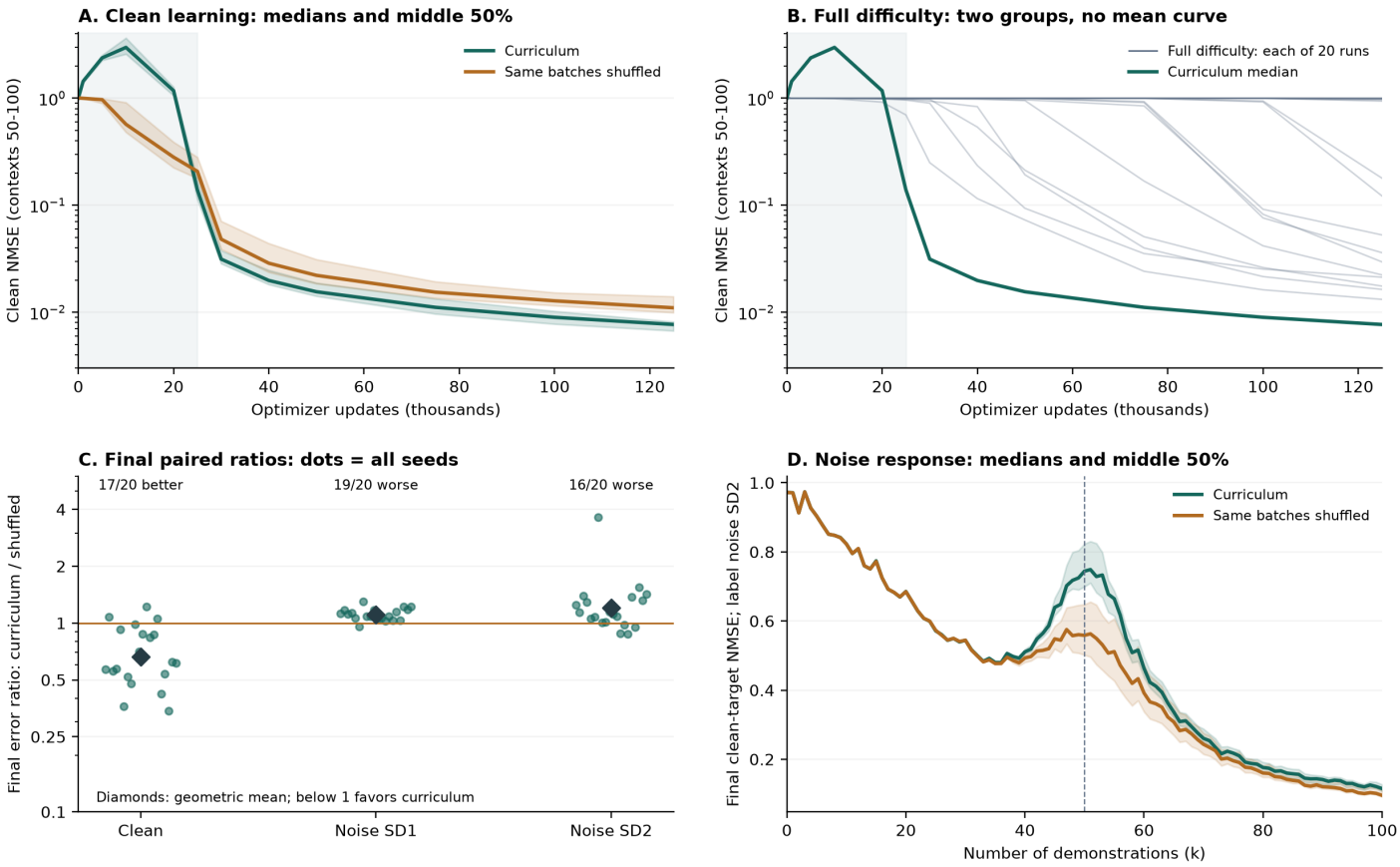


Figure. A: median clean curves with interquartile bands. B: all full-difficulty runs against the curriculum median; shading marks the first 25k updates. C: paired final ratios for every seed, with geometric means; values below 1 favor curriculum. D: noisy SD2 median error curves and interquartile bands; the dashed line marks k = 50.

A mechanism hypothesis, not a demonstrated explanation

With SD2 noise, curriculum has a sharper error peak near k = d = 50. An exploratory ratio of mean error at k = 48-52 to k = 38-42 is larger in 16/20 seeds (exact Wilcoxon p = 0.000395); median ratios are 1.431 versus 1.127. At SD1 it is larger in 19/20 (p = 0.00000572), but both median ratios are below 1 (0.815 vs 0.678), so this alone indicates less decline, not a local peak. The pattern is consistent with weaker implicit regularization; it does not demonstrate an OLS-like algorithm. Fragile post hoc ridge fits are omitted. Independent reference predictions and a fresh replication are needed.

Positioning and background reuse

Garg et al. already showed curriculum benefits for synthetic ICL [1]; Liu et al. study task complexity, architecture and curricula [2]. The closest ordering work is Shin & Alvarez-Melis [3], whose static control matches cumulative exposure by difficulty level. Our exact realized-batch reshuffling in regression and the clean/noisy trade-off are potential contributions, rather than an established first-in-literature claim. One task family, architecture and optimizer grid do not establish generality.

Reuse the original ICL definitions and regression motivation [4], while preserving its negative noise-ordering result. **Correction:** the inspected synthetic repository already averages loss over all prediction positions. The earlier packet inferred final-query-only training from the paper text; that inference was wrong. Reconcile the original text with historical run provenance before revising its methods.

[1] Garg et al. (2022), *What Can Transformers Learn In-Context?* arxiv.org/abs/2208.01066
[2] Liu et al. (2025), *Probing In-Context Learning: Impact of Task Complexity and Model Architecture on Generalization and Efficiency.* arxiv.org/abs/2505.06475
[3] Shin & Alvarez-Melis (2026), *Curriculum Learning as Transport: Understanding Curricula with Wasserstein Geodesics.* arxiv.org/abs/2609.09099
[4] Gadipudi, Pattanaik, Syrdal & Vestrum, *Robustness of In-context Learning via Curriculum Learning.* Original project PDF